S.no	Two Stroke Engine	Four Stroke Engine
13.	These engines are easy to manufacture.	These engines are comparatively hard to manufacture.
14.	More wear and tear occurs	Less wear and tear occurs.
15.	A part of air fresh air fuel mixture mixes with the exhaust gas, hence less power.	Burnt out gases is not mixed with fresh charge, hence more power.
16.	Fuel consumption is more.	Fuel consumption is less
17.	High power-to-weight ratio	Less power-to-weight ratio
18.	More noise	Less noise
19.	Air cooled engine.	Air / water cooled engine.
20.	Works based on Clerk cycle.	Works based on otto / diesel cycle
21.	Easy lubrication due to lubrication oil mix with the fuel.	Complex lubrication mechanism
22.	Consumption of lubrication oil is more because some oil burns with fuel.	Consumption of lubrication oil is less
23.	Less weight	More weight
24.	Used in scooter, mopeds, auto rickwsaw.	Used in motor cycles, autoricksaw, car, bus, lorry, trucks, tractors etc.

5.7 FOUR STROKE DIESEL ENGINE

The engine which uses diesel as a fuel is called as diesel engine. The compression ratio for diesel engine is higher and it will be ranging from 16 : 1 to 20 : 1. The compression ratio for a petrol engine is 10:1. During suction stroke, the air alone is inducted. The fuel is pressured and distributed to various cylinders through the fuel injection pump. Diesel injector will be located in cylinder head for injecting the diesel fuel at high pressure. There is no spark plug. All the remaining parts are similar to petrol engine.

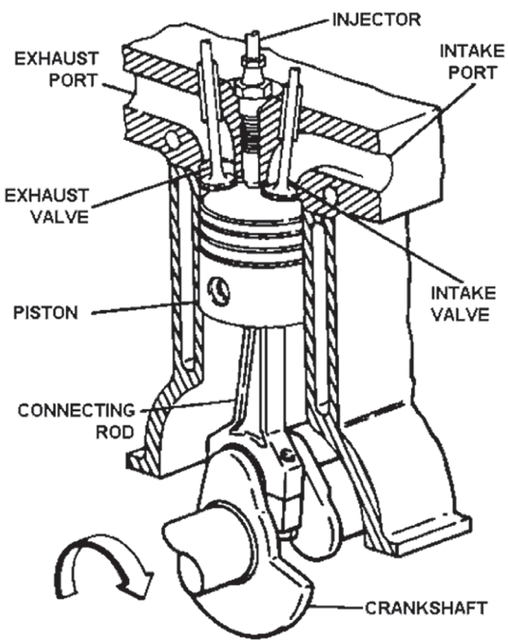


Figure 5.7(a) Four Stroke Diesel Engine

Construction

The construction of a single cylinder diesel engine is shown in above figure. The reciprocating motion of a piston is converted into rotary motion of a crankshaft through connecting rod. The piston and the crankshaft are connected by means of a connecting rod. The flywheel is attached to the one end of the

AM_Unit_05.indd 105 12/20/2019 5:26:39 PM